

3 Adapter Same Dataset Merging

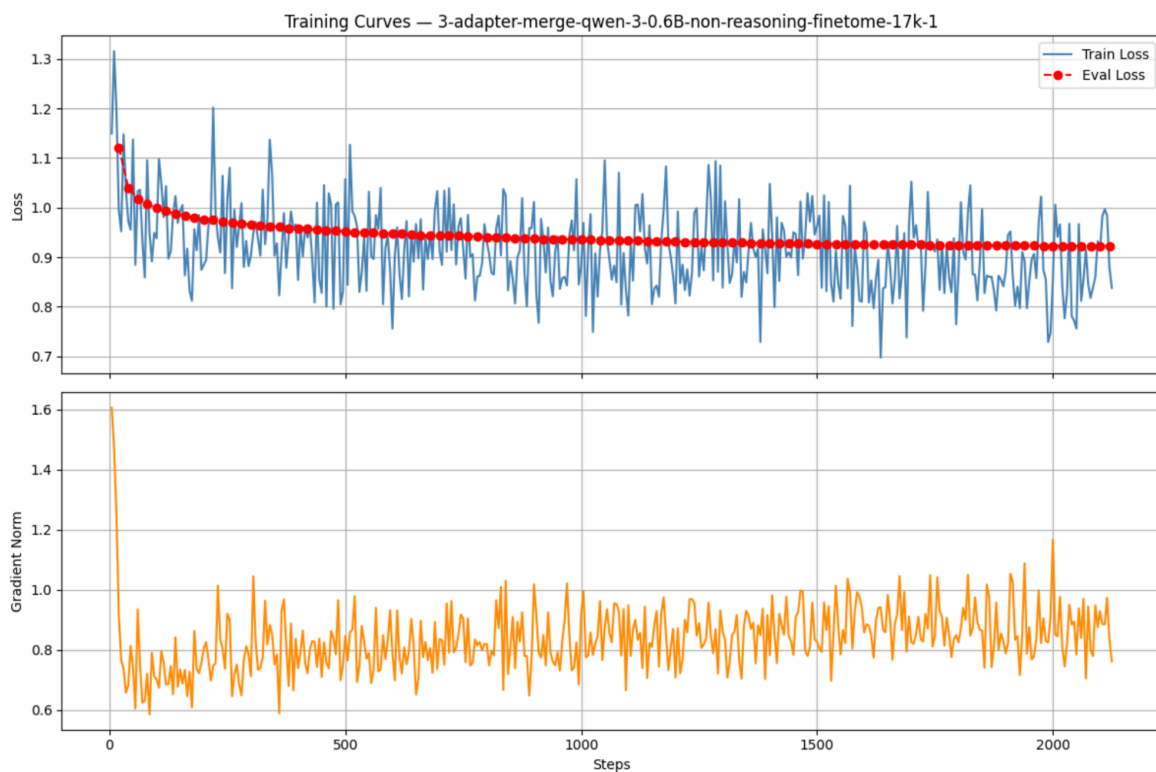
1) Training:

Model used: Qwen 3 0.6B

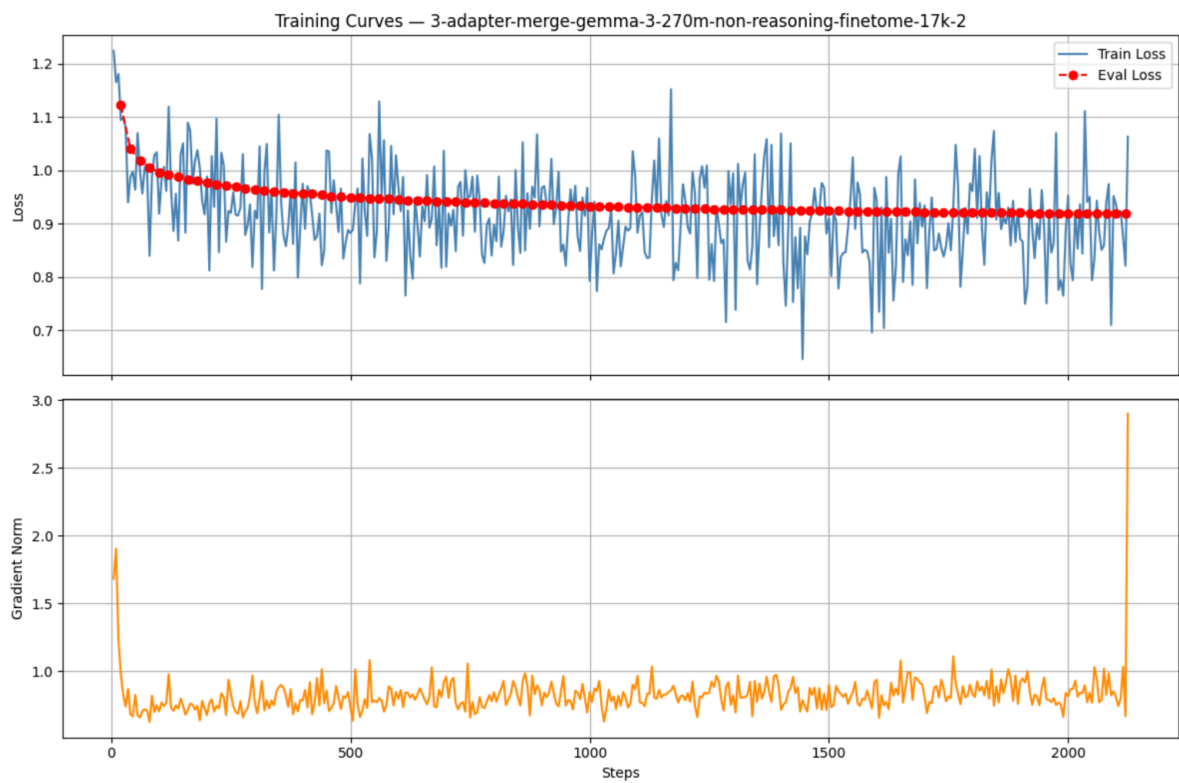
Dataset used: Finetome-100k

- Only 51K samples of the dataset were used for training
- 3 Models were trained on 17K subsets of the dataset
- 1 Model was trained on 51K samples of the dataset.

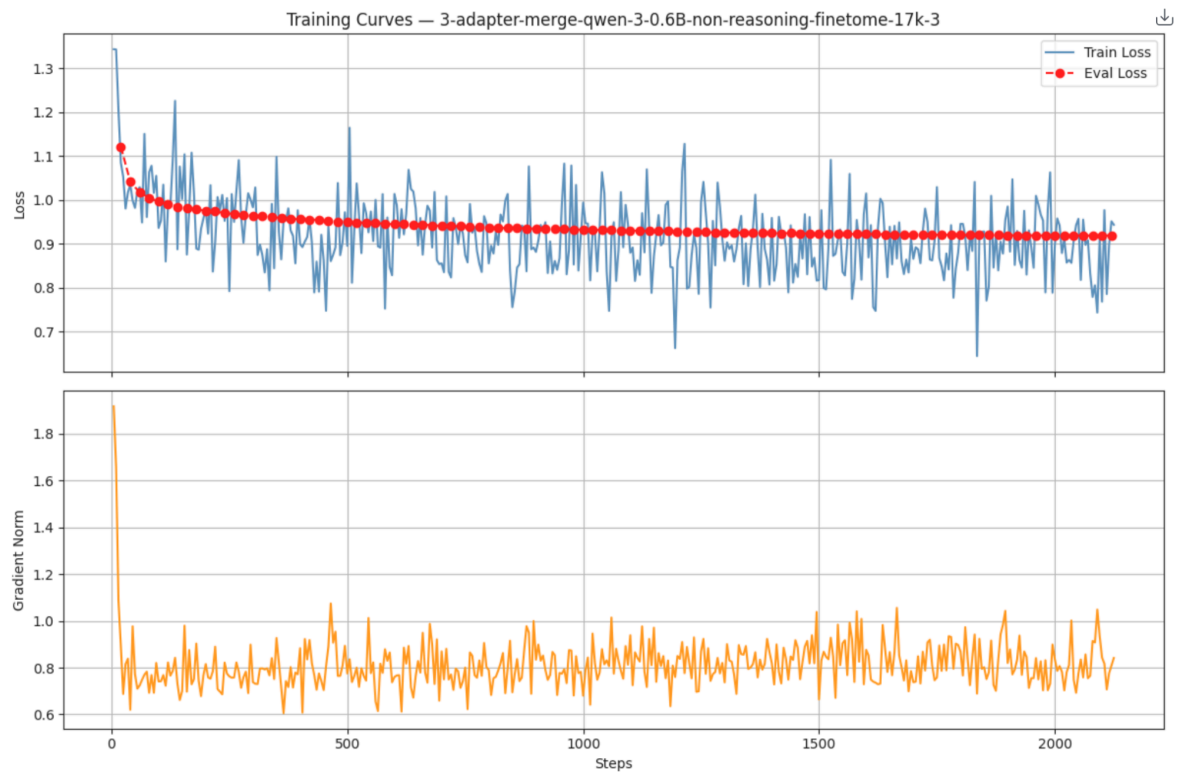
Training loss curve for adapter that was trained on first subset



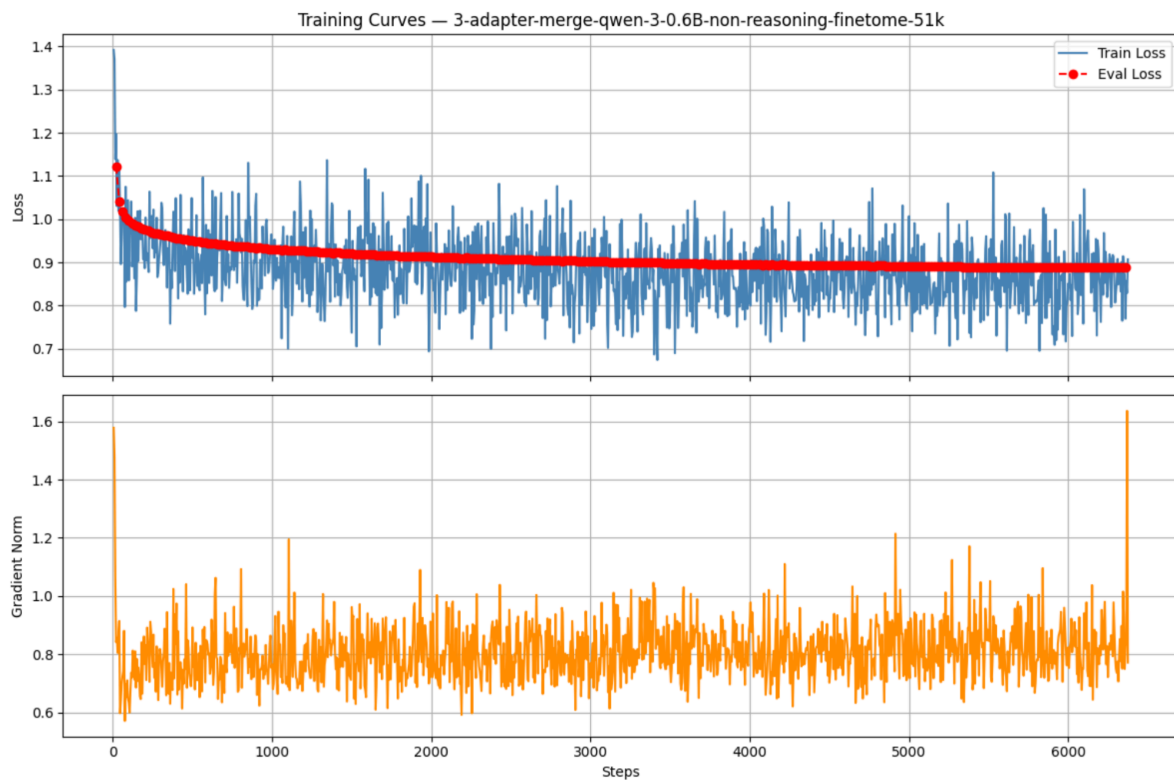
Training loss curve for adapter that was trained on second subset:



Training loss curve for adapter that was trained on third subset:



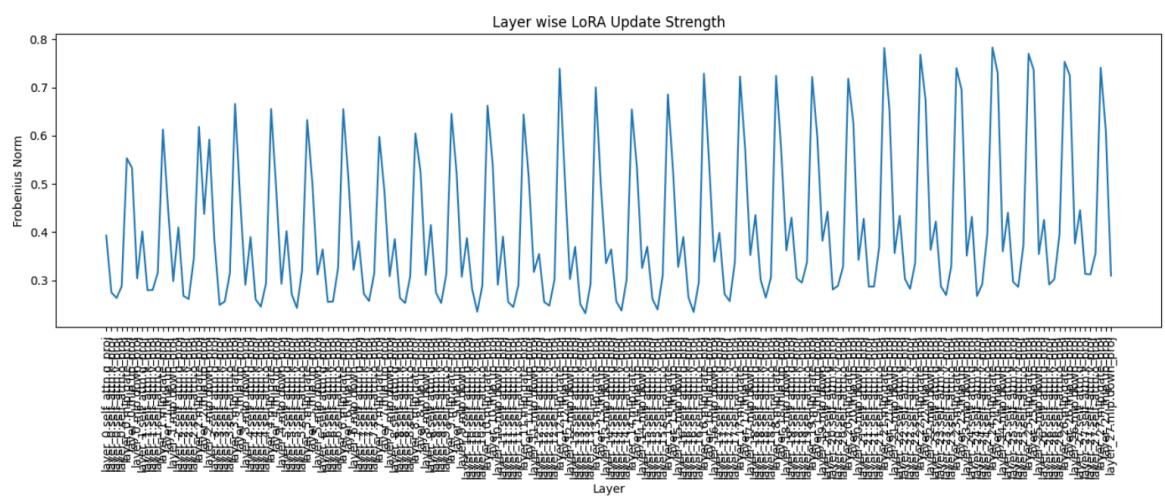
Training loss curve for adapter that was trained on whole set:



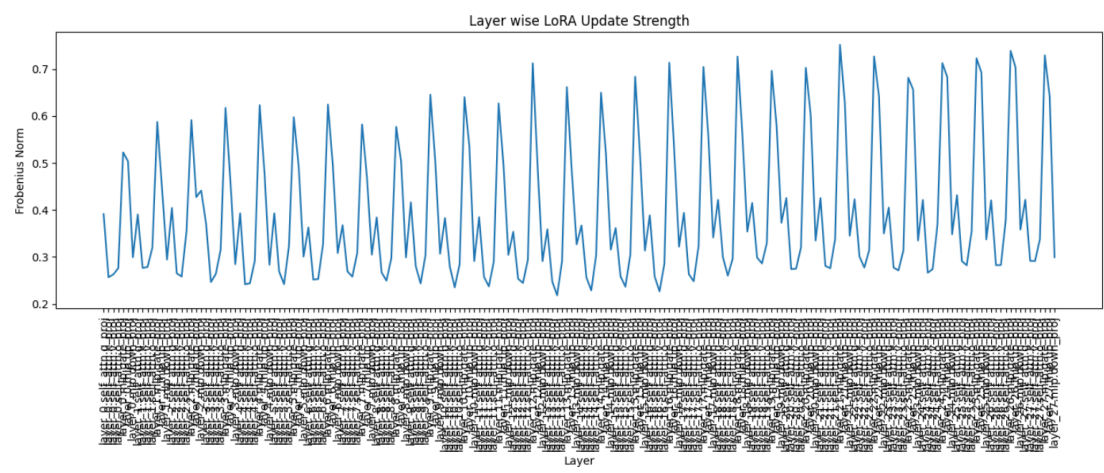
2) Comparison:

2.1) Frobenius norm of

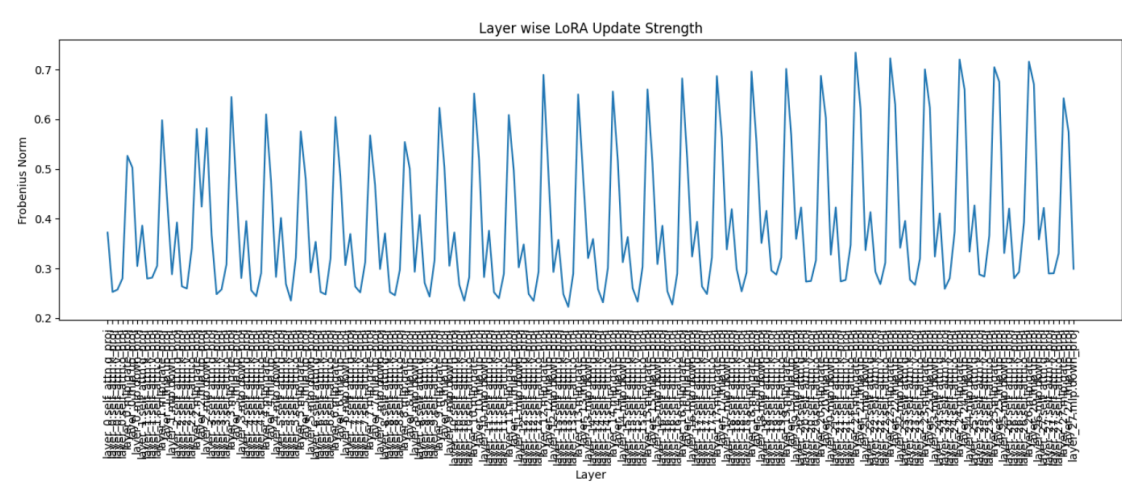
- Model 1:



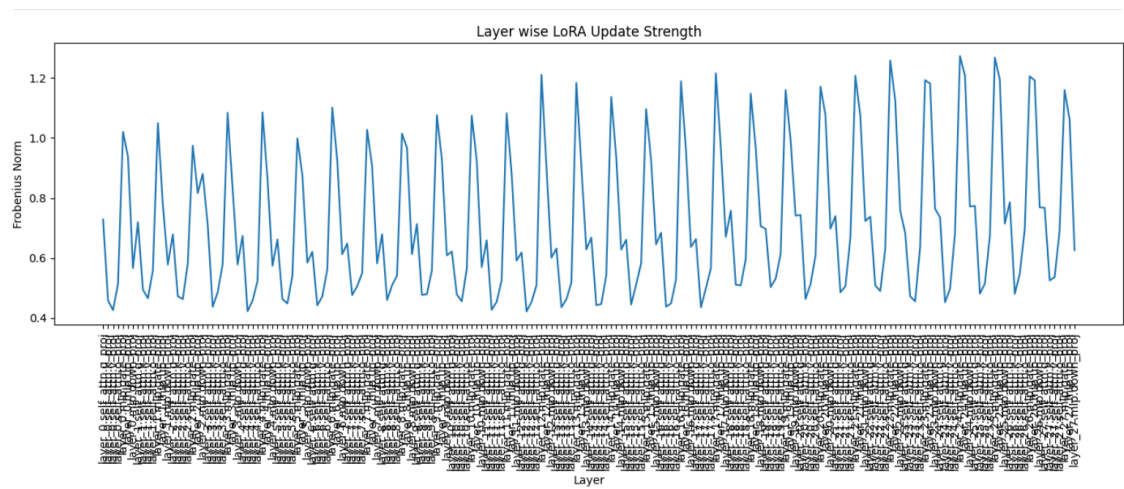
- **Model 2:**



- **Model 3:**



- **Full model**



2.2) Cosine similarity

The moderate cosine similarity between subset-trained and full-model adapters (~0.62–0.73) indicates that each 17K subset captured a meaningfully distinct but overlapping subspace of the full 51K gradient signal. This is the prerequisite for merging to be non-trivial — if similarity were near 1.0, merging would be redundant; near 0, it would be destructive.

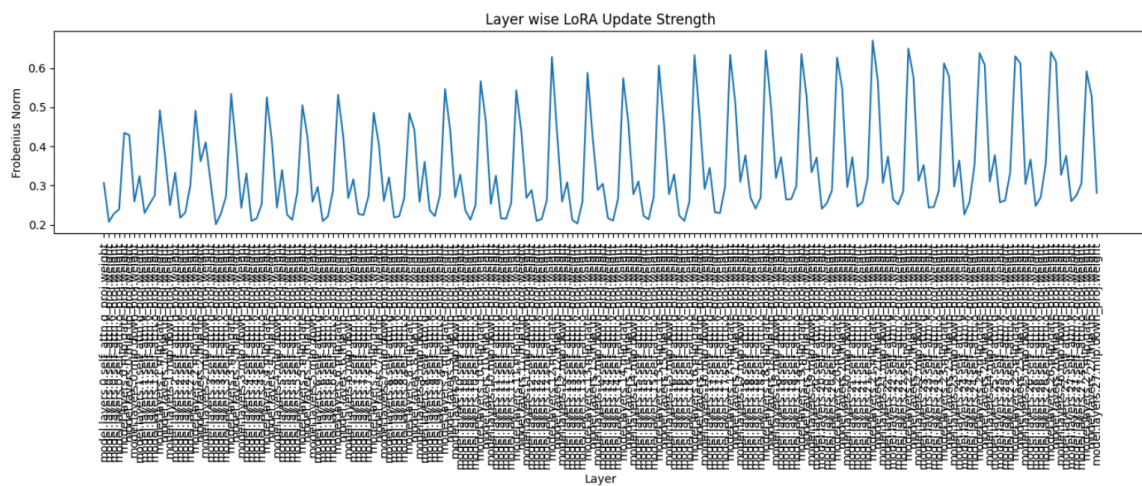
layer	cosine_similarity
layer_0.mlp.down_proj	0.648002
layer_0.mlp.gate_proj	0.624881
layer_0.mlp.up_proj	0.665026
layer_0.self_attn.k_proj	0.593485
layer_0.self_attn.o_proj	0.635905
layer_0.self_attn.q_proj	0.620560
layer_0.self_attn.v_proj	0.687272
layer_1.mlp.down_proj	0.640153
layer_1.mlp.gate_proj	0.637818
layer_1.mlp.up_proj	0.666485
layer_1.self_attn.k_proj	0.631054
layer_1.self_attn.o_proj	0.653132
layer_1.self_attn.q_proj	0.629811
layer_1.self_attn.v_proj	0.712945
layer_10.mlp.down_proj	0.665839
layer_10.mlp.gate_proj	0.693702
layer_10.mlp.up_proj	0.716765
layer_10.self_attn.k_proj	0.699113
layer_10.self_attn.o_proj	0.673173
layer_10.self_attn.q_proj	0.687808
layer_10.self_attn.v_proj	0.731733

3) Merging

3.1) Linear

$$\Delta W_{\text{merged}} = \sum_i w_i \cdot \Delta W_i \quad \text{where } w_i = 1/n$$

$$\theta^* = \theta_0 + (1/3) \cdot \Delta W_1 + (1/3) \cdot \Delta W_2 + (1/3) \cdot \Delta W_3$$



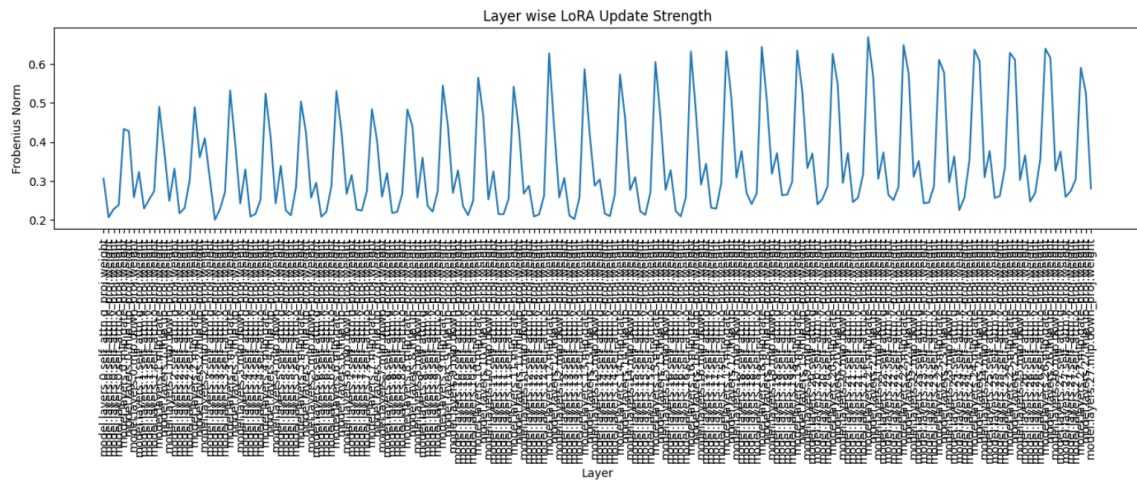
3.2) SVD

$$\Delta W_{\text{avg}} = \sum_i w_i \cdot \Delta W_i \quad \leftarrow \text{linear combine}$$

$$U, S, V^T = \text{SVD}(\Delta W_{\text{avg}})$$

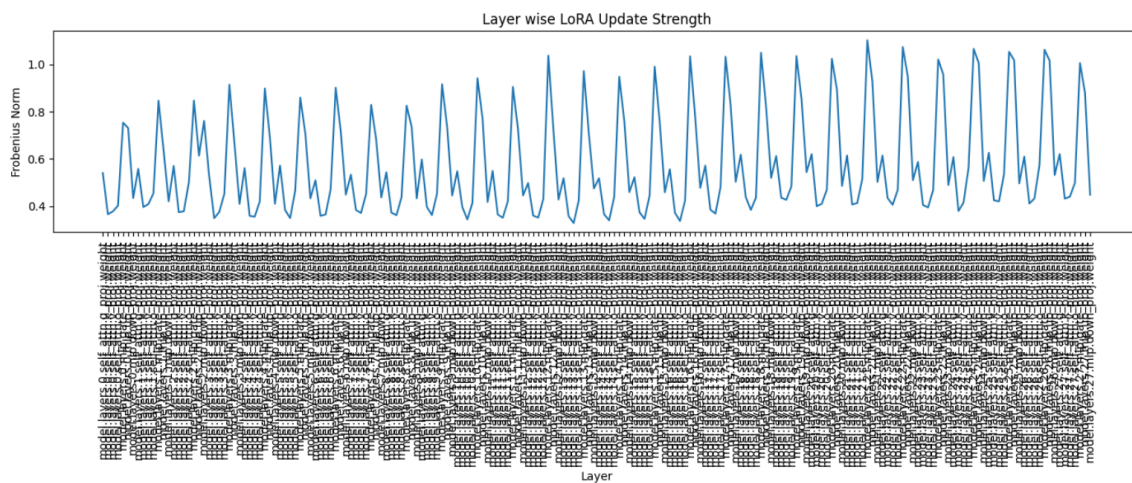
$$\Delta W_{\text{merged}} = U[:, :r] \cdot \text{diag}(S[:, r]) \cdot V^T[:, :] \quad \leftarrow \text{truncate to rank } r=16$$

$$\theta^* = \theta_0 + \Delta W_{\text{merged}}$$



3.3) DARE

Randomly drops 80% of each adapter's parameters with probability $(1-p)$, then rescales the survivors by $1/p$ to preserve the expected magnitude. Reduces redundancy and overfitting to any single adapter's quirks. Seeded for reproducibility.



3.4) TIES

$\tau_i = \text{trim}(\Delta W_i, \rho=0.2)$ $\leftarrow$ keep top 20% by $|\Delta W|$

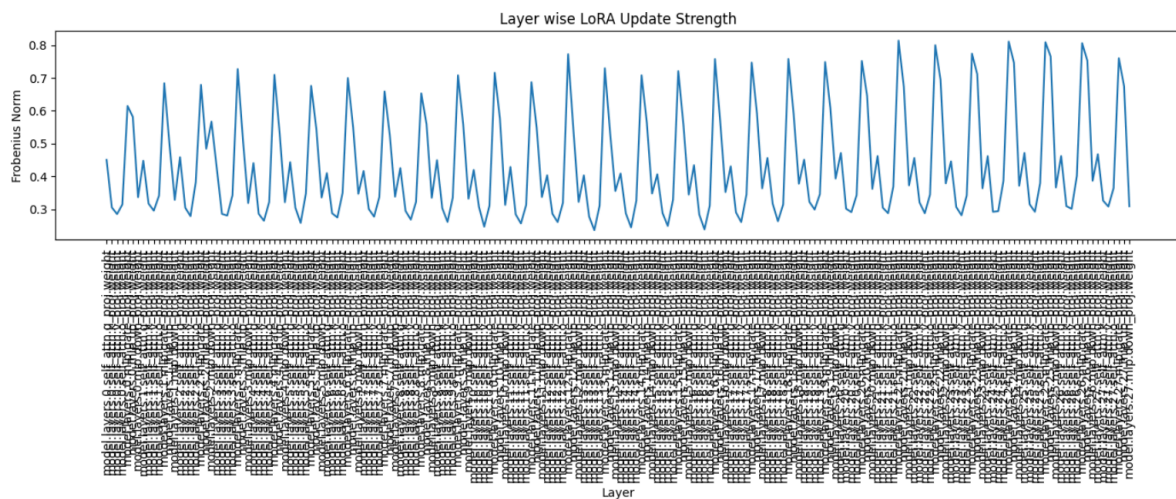
$\gamma = \text{sign}(\sum_i \tau_i)$ $\leftarrow$ majority vote on sign

$m_i = \mathbb{1}[\text{sign}(\tau_i) = \gamma]$ $\leftarrow$ agreement mask

$$\Delta W_{\text{merged}} = (\sum_i w_i \cdot \tau_i \cdot m_i) / (\sum_i w_i \cdot m_i + \epsilon)$$

$$\theta^* = \theta_0 + \Delta W_{\text{merged}}$$

Designed to resolve sign conflicts between adapters. Parameters where adapters disagree on direction are excluded from the merge, preventing destructive interference.



3.5) SLERP

$\hat{u}_1 = \Delta W_1 / \|\Delta W_1\|$ $\hat{u}_2 = \Delta W_2 / \|\Delta W_2\|$ $\leftarrow$ unit vectors

$\Omega = \arccos(\text{clamp}(\hat{u}_1 \cdot \hat{u}_2, -1+\epsilon, 1-\epsilon))$ $\leftarrow$ angle between

$\text{SLERP}(\Delta W_1, \Delta W_2, t) =$

$$[\sin((1-t)\Omega)/\sin(\Omega)] \cdot \hat{u}_1 + [\sin(t\Omega)/\sin(\Omega)] \cdot \hat{u}_2 \times \text{llinterp}$$

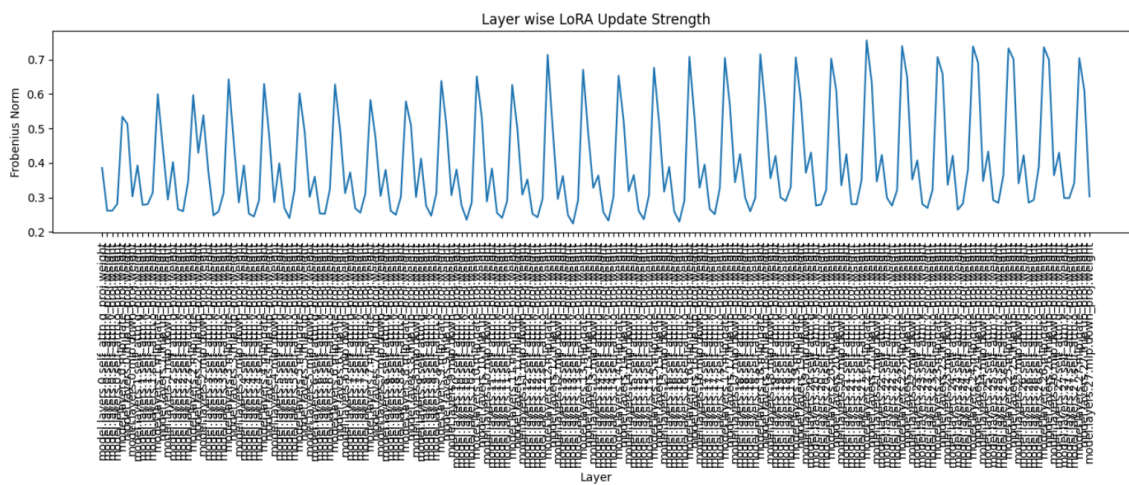
For n=3 adapters, applied sequentially with $t = [0.5, 0.5]$:

$$\Delta W_{\text{tmp}} = \text{SLERP}(\Delta W_1, \Delta W_2, 0.5)$$

$$\Delta W_{\text{merged}} = \text{SLERP}(\Delta W_{\text{tmp}}, \Delta W_3, 0.5)$$

$$\theta^* = \theta_0 + \Delta W_{\text{merged}}$$

Interpolates along the surface of a hypersphere rather than a straight line through weight space. Preserves the magnitude of each delta and avoids the collapse that happens when linearly averaging nearly-orthogonal vectors. For 3 adapters, merging is done sequentially — first two adapters are merged, then the result is merged with the third



4) Evaluation

MODEL	R1	R2	RL
Qwen3-0.6B-linear-3-adapters-same-dataset-merged	0.3416	0.0832	0.1673
Qwen3-0.6B-svd-3-adapters-same-dataset-merged	0.3368	0.0820	0.1667
Qwen3-0.6B-dare-3-adapters-same-dataset-merged	0.3278	0.0778	0.1626
Qwen3-0.6B-ties-3-adapters-same-dataset-merged	0.3361	0.0836	0.1673
Qwen3-0.6B-slerp-3-adapters-same-dataset-merged	0.3370	0.0846	0.1676
full_finetome	0.3389	0.0872	0.1703

Key Observations:

- 1. Linear merging achieves the highest ROUGE-1 score (0.3416), marginally exceeding the full model (0.3389).**

Three adapters each trained on disjoint 17K subsets, when uniformly averaged, recover and slightly surpass the unigram overlap of a model trained on the full 51K corpus. Training several models independently on random partitions and then combining them (ensemble/merge) can outperform a single model because the independent models average out optimization noise and reduce variance. A single full-data model follows only one SGD trajectory and may be more affected by the ordering of training examples, giving some implicit preference to examples seen later in training

2. **SLERP and TIES perform comparably to Linear on ROUGE-1 and ROUGE-L (0.3370, 0.3361 vs 0.3416), suggesting that sign conflicts and magnitude collapse are minimal when adapters share the same training distribution.** TIES's sign-agreement filtering and SLERP's spherical interpolation are both designed to handle adapter disagreement — their marginal gains over Linear are limited here precisely because the same-dataset setting produces highly consistent adapter directions, leaving little destructive interference to resolve.
3. **SVD merging closely tracks Linear (ROUGE-1: 0.3368, ROUGE-L: 0.1667).** The rank-16 truncation incurs negligible information loss, which is consistent with the effective rank analysis showing adapter deltas concentrate their energy in approximately 10–13 singular components. Truncating to the nominal LoRA rank therefore discards near-zero singular values rather than live signal.
4. **DARE underperforms all other methods across all metrics (ROUGE-1: 0.3278, ROUGE-2: 0.0778, ROUGE-L: 0.1626).** DARE's random parameter dropping at $p=0.8$ is designed to eliminate redundancy in cross-task merging scenarios where parameters serving different tasks may constructively interfere. In this same-dataset setting, however, parameters that DARE drops are not conflicting but consistently informative across all three adapters — the sparsification therefore discards genuine signal rather than noise, leading to measurable degradation.

5. The full model achieves the highest ROUGE-2 score (0.0872). ROUGE-2 measures bigram overlap and is sensitive to local fluency and phrasal coherence. That the full model retains an advantage here despite losing on ROUGE-1 suggests that sequential exposure to 51K examples produces stronger bigram co-occurrence statistics than what can be recovered by merging three independently trained adapters. Three disjoint 17K training runs do not replicate the full data's sequential co-occurrence signal, and this gap manifests specifically at the n-gram level.